Cleaned_Omair_Sales_Data.csv

InsightFlow AI analysis report · Sales dataset · generated August 05, 2026

Executive Summary

This sales dataset reflects a total revenue of approximately 10 million dollars across nearly 2,000 transactions, revealing a consistent year-end demand surge that highlights a clear opportunity for seasonal inventory optimization. While the overall data quality is exceptionally high at 99.96%, significant historical gaps between 2006 and 2015 render long-term trend analysis unreliable for strategic forecasting. Moving forward, leadership should address the strong reliance on cost-plus pricing, which currently limits our flexibility to adjust margins during periods of cost volatility.

Key Metrics

Metric	Value
Total Revenue	10075360.98$
Total Orders	1998
Average Transaction Value	5042.72$
Data Quality Score	99.96%

Insights

High Cost-to-Sale Correlation [INFO]
The strong 0.86 correlation between Cost and Sale suggests that pricing models are tightly tethered to input costs. This indicates a potential lack of pricing power or a cost-plus strategy, which may be limiting profit margins during periods of high cost volatility.

Cyclical Sales Volatility [INFO]
Data shows consistent peaks in performance during December, with sales hitting 493,536 in Dec 2004 and 477,993 in Dec 2005. This seasonality suggests that resource allocation and inventory stocking should be optimized ahead of year-end to capitalize on this repeatable demand surge.

Significant Data Gaps Impacting Longitudinal Analysis [CRITICAL]
There is a 98-month data gap between 2006-06 and 2014-08, followed by a 16-month gap leading to 2015-12. These breaks render any analysis of long-term business trajectory invalid, as the recent records (only 3 total since 2014) lack the historical context needed to form a reliable trend.

Presence of Cost Outliers [WARNING]
The presence of 5 cost outliers in a 1,998-record dataset represents a minor risk to the reliability of cost-based profitability metrics. It is worth auditing these specific transactions to determine if they are legitimate high-cost business operations or entry errors that could skew regional margin analysis.

Data Cleaning

• **Cost:** 5 IQR-based outlier value(s) found and left in place (not removed automatically).

Forecast

Not enough complete time periods (need at least 4, found 1) after excluding 26 earlier period(s) separated from the recent data by a large gap